

functions. This platform allows the implementation of all these functionalities by providing configuration management features that work with any arbitrary data structure. Hence we can even apply the configuration data that's as simple as a key-value set or as complex as nested objects. Kaa constantly tracks already applied configuration data as well as pending delivery, as different IoT devices may not be constantly connected. Kaa supports both push and pull modes as the default configuration management protocol. To ensure reliability, configuration delivery is completely based on the result codes and configuration application confirmations. Kaa generates a config change event whenever the device changes its current configuration. These generated events can be used to trigger different automation processes inside as well as outside the platform. The Kaa platform allows us to manage the configuration of its devices, individually. We can define the configuration for a specific device or even for a group of devices on the basis of their individual characteristics like location, software version, etc.

8. **Supports over-the-air updates:** Over-the-air updates is a powerful feature that enables devices to continuously evolve over time. We can implement many new features for the devices already in use—for instance, auto pilots for vehicles, face recognition for security cameras, predictive maintenance for industrial machineries, etc. Kaa allows reliable delivery of software updates by making use of the confirmation response codes that are sent by devices once the results are updated. Another significant feature is its ability to define the software update path by using

compatibility graphs based semantic versions. Kaa also allows tracking of the current version of software that is installed on the managed devices and helps in defining its potential update vectors. Each software version is represented in Kaa using a flexible descriptor that's delivered to the device in the form of an instruction to perform an update.

A quick overview of the architecture of Kaa

The Kaa IoT platform has been developed keeping heavy-duty, enterprise-grade IoT solutions in mind. It completely removes the monolithic approach implemented while developing an architecture that supports highly portable microservices, hence allowing flexible customisation and rearrangement even in the midst of the solution's life cycle. It also enables cloud-native scalability while offering full freedom with regard to the type of deployment. The Kaa IoT platform consists of Kaa extensions, the Kaa server and the endpoint SDKs as the main components.

1. Kaa extensions are nothing but the independent software modules that improve the functionality of the platform.
2. The Kaa server is actually the back-end part of the platform. It manages tenants, users' applications and devices. It also exposes the integration interfaces and provides administrative capabilities.
3. Endpoint SDKs are the sets of libraries that offer client side APIs for different features of the Kaa platform and handle data marshalling, communication, persistence, etc. Kaa SDKs are designed to facilitate the development of client applications that need to be run on different connected devices. There are various endpoint SDK types that are available in different programming languages.

Security in Kaa

The communication of Kaa with different devices connected to it is secured with TLS or DTLS, by default. Security is of course one of the main parameters for all enterprises when they opt for a platform. In the case of Kaa, various devices connected to it authenticate the server by validating its TLS certificate. The Kaa server supports a couple of methods for

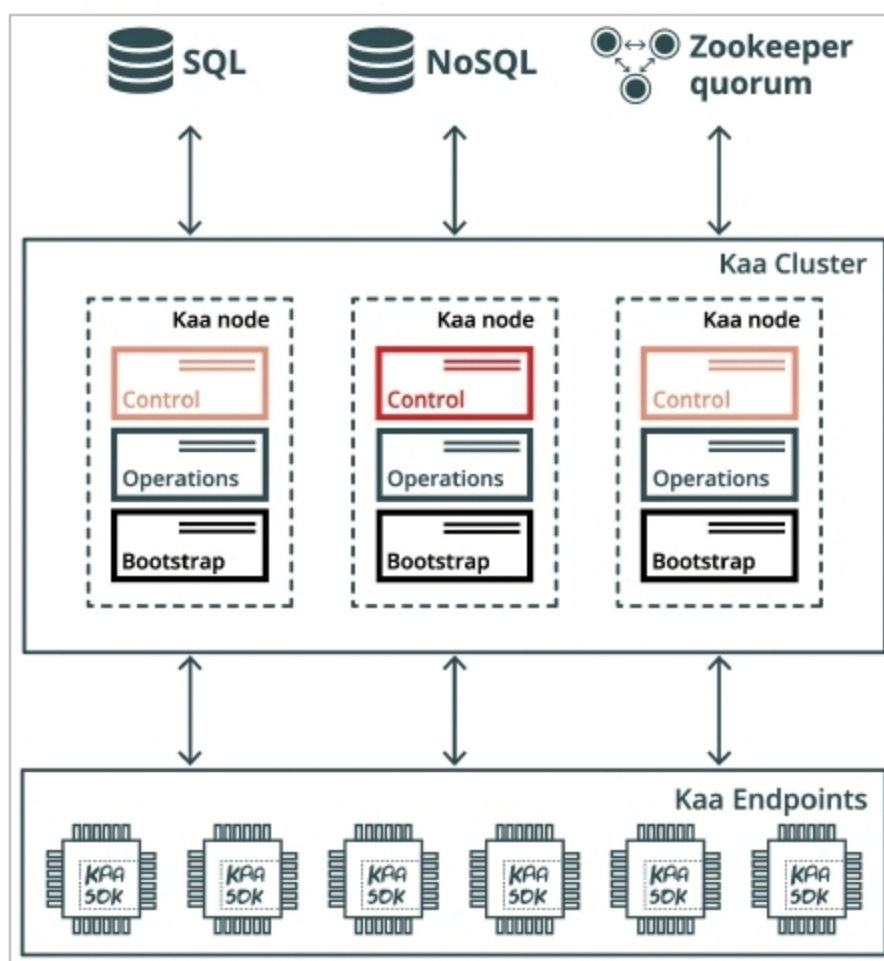


Figure 2: Architecture of Kaa (Image source: [googleimages.com](https://www.googleimages.com))



Figure 3: Kaa — middleware that connects hardware and software applications (Image source: [googleimages.com](https://www.googleimages.com))